

The empirical recurrence for OEIS A305171, recovered from its tail

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Abstract

OEIS A305171 records “Empirical recurrence of order 66” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 70$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 68$. Its last coefficient is $c_{66} = 128 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A305171 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 2, 3, 4, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

0, 18, 46, 410, 2397, 13970, 93426, 586718, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (May 26 2018 18:15:04, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 70$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 70$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 140$ exact terms from $n = 2$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	4	c_{18}	−138567	c_{35}	51550824	c_{52}	−3588533
c_2	16	c_{19}	−290204	c_{36}	51620134	c_{53}	6806202
c_3	39	c_{20}	−101924	c_{37}	48742490	c_{54}	4654327
c_4	−212	c_{21}	3078792	c_{38}	−22502647	c_{55}	−962904
c_5	−624	c_{22}	7053091	c_{39}	−158998476	c_{56}	−4366913
c_6	28	c_{23}	3642662	c_{40}	−56208441	c_{57}	−807691
c_7	3174	c_{24}	2529923	c_{41}	30263257	c_{58}	1085745
c_8	5251	c_{25}	10004756	c_{42}	−126113951	c_{59}	364022
c_9	−3013	c_{26}	2653637	c_{43}	−7137742	c_{60}	−75786
c_{10}	−16363	c_{27}	−503555	c_{44}	8063014	c_{61}	−17532
c_{11}	−12304	c_{28}	58827924	c_{45}	−92816251	c_{62}	31207
c_{12}	280	c_{29}	78183406	c_{46}	−16484316	c_{63}	1514
c_{13}	63481	c_{30}	6986981	c_{47}	60527306	c_{64}	−4820
c_{14}	142704	c_{31}	48645815	c_{48}	−22329134	c_{65}	−552
c_{15}	−130896	c_{32}	36378302	c_{49}	−12237125	c_{66}	128
c_{16}	−500954	c_{33}	−47429889	c_{50}	29881896		
c_{17}	−430816	c_{34}	35803789	c_{51}	−4900809		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 68$, and it is the recurrence OEIS A305171 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{65} - \dots - c_{66}$. Its constant term is $-c_{66} = -128$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 128.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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